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Sensing the World

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# Final Project Reflection

Over three class periods, our team (Diego, Diane, and I) built an automated floodgate prototype designed to detect rising water levels and humidity to protect vulnerable areas from flash flooding. We divided our responsibilities based on our strengths. I focused heavily on the structural design and physical construction, while Diego and Diane handled writing the code, refining the circuit wiring, and integrating the electronic components. The structure was built using cardboard and wooden popsicle sticks, incorporating a top-mounted pulley system driven by a stepper motor to raise and lower the gate door smoothly. For sensors, we wired a standard rain sensor for direct liquid contact along with a PEDOT sensor to measure ambient humidity changes. We also wired status LEDs and a push button so the gate could be manually reset or toggled when the conditions were dry. To speed up software development, we used AI to draft our initial Python test scripts for the stepper motor, rain sensor, and humidity sensor before integrating everything into the final control loop.

Working on the physical build gave me a real appreciation for how the mechanics affect electronic control. Even though the logic code was straightforward, getting the stepper motor to lift the gate required precise guide alignment and consistent string tension across the pulleys so the door wouldn't bind in the cardboard tracks. Dividing up the workload made us a lot more productive. For instance, I could concentrate on re-engineering the frame and track guides while

my teammates debugged motor driver issues on the breadboard. When we finally tested the completed build, wetting either sensor instantly dropped the gate and flipped the indicator light from green to red. If we were to take this prototype further, I would rebuild the cardboard frame out of laser-cut acrylic or 3D-printed plastic for better waterproofing, and we would integrate a dedicated, clean power supply rather than running everything off breadboard jumpers. Overall, seeing our structural mechanism and the electronics work together seamlessly made the whole project a great experience.